

Antu Saha

Ph.D. Candidate, Department of Computer Science, William & Mary, Williamsburg, VA, USA

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RESEARCH INTERESTS

My research lies at the intersection of **AI for Software Engineering** and **Software Maintenance and Evolution**. I design, build, and evaluate techniques that leverage **large language models (LLMs)**, **multimodal learning**, and **empirical methods** to automate software quality tasks—particularly bug report management, issue resolution, and developer tooling. My work bridges automated techniques and real-world practice through human studies, developer interviews, and large-scale dataset construction.

EDUCATION

William & Mary, Virginia, USA

Ph.D., Computer Science

Jan 2022 – May 2027 (expected)

- Dissertation: *Studying and Automating Issue Resolution for Software Quality*
- Advisor: **Dr. Oscar Chaparro**
- CGPA: 3.96/4.00

Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh

B.Sc., Computer Science & Engineering

Mar 2013 – Dec 2017

- **3rd among 120 students** in the class (CGPA: 3.82/4.0)

PUBLICATIONS

Manuscripts Under Review

- [1.] **A. Saha**, S. Pandey, A. Marcus, K. Moran, and O. Chaparro, “**What Bug Report Information Helps Agents Fix UI-Centric Bugs? An Empirical Study of Android UI Bug Repair**,” 2026. [Research Track]
- [2.] **A. Saha**, S. Bennett, and O. Chaparro, “**BURT++: An Interactive AI-Powered Bug Reporting System for Mobile Applications**,” 2026. [Research Track]
- [3.] **A. Saha**, A. Dipongkor, S. Bennett, K. Moran, A. Marcus, and O. Chaparro, “**Automatically Enhancing the Quality of Android App Bug Reports**,” *arXiv preprint arXiv:2604.01148*, 2026. [Research Track]
- [4.] M. Sun, **A. Saha**, N. Silva, A. Mastropaolo, and O. Chaparro, “**Fine-grained Multi-Document Extraction and Generation of Code Change Rationale**,” *arXiv preprint arXiv:2604.10345*, 2026. [Research Track]
- [5.] **A. Saha**, M. Sun, and O. Chaparro, “**Automatically Identifying Solution-Related Content in Issue Report Discussions with Language Models**,” *arXiv preprint arXiv:2511.06501*, 2025. [Research Track]

Refereed Publications

- [6.] [ICSME’26] A. Adnan*, M. Rahman*, **A. Saha**, and O. Chaparro, “**Rethinking Issue Resolution for AI/ML Systems**,” in *Proceedings of the 42nd International Conference on Software Maintenance and Evolution*, 2026. (**M.Sc. students I mentored*) [Vision Track]
- [7.] [ICSE’26] **A. Saha**, “**Studying and Automating Issue Resolution for Software Quality**,” in *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering*, 2026. [Doctoral Symposium Track]
- [8.] [ICSE’25] **A. Saha** and O. Chaparro, “**Decoding the Issue Resolution Process in Practice via Issue Report Analysis: A Case Study of Firefox**,” in *Proceedings of the 47th IEEE/ACM International Conference on Software Engineering*, 2025. [Research Track]
- [9.] [ICPC’25] J. Mahmud*, **A. Saha***, O. Chaparro, K. Moran, and A. Marcus, “**Combining Language and App UI Analysis for the Automated Assessment of Bug Reproduction Steps**,” in *Proceedings of the 33rd IEEE/ACM International Conference on Program Comprehension*, 2025. (**co-first author: equal contribution*) [Research Track] (41% acceptance rate)
- [10.] [MSR’25] A. Adnan*, **A. Saha**, and O. Chaparro, “**Sprint: An Assistant for Issue Report Management**,” in *Proceedings of the 22nd IEEE/ACM International Conference on Mining Software Repositories*, 2025. (**M.Sc. student I mentored*) [Tool Demo Track]

- [11.] [\[ISSTA'24\]](#) A. Saha, Y. Song, J. Mahmud, Y. Zhou, K. Moran, and O. Chaparro, “**Toward the Automated Localization of Buggy Mobile App UIs from Bug Descriptions**,” in *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis*, 2024, pp. 1249–1261. **[Research Track] (21% acceptance rate)**
- [12.] [\[IJEECS\]](#) M. Begum, B. C. Das, M. Z. Hossain, A. Saha, and K. A. Papry, “**An Improved Kohonen Self-Organizing Map Clustering Algorithm for High-Dimensional Data Sets**,” in *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 24, Oct. 2021.
- [13.] [\[ECCE'19\]](#) A. Saha, J. Saha, and B. Sen, “**An Expert Multi-Modal Person Authentication System Based on Feature Level Fusion of Iris and Retina Recognition**,” in *Proceedings of the 2019 International Conference on Electrical, Computer and Communication Engineering*, 2019, pp. 1–5.

RESEARCH EXPERIENCE

William & Mary, Virginia, USA

- **Graduate Research Assistant** Jan 2022 – Present
 - Built a suite of **LLM-powered bug report tools**—AstroBR [\[ICPC'25\]](#), BugScribe, and BURT++—that leverage **app context** (e.g., GUI screenshots, interaction traces) to automatically assess, enhance, and interactively guide the creation of mobile app bug reports; designed **structured evaluation rubrics** and an **LLM-as-a-judge** framework validated through **human studies**
 - Designed and executed large-scale experiments comparing **ML models** (e.g., Decision Trees, Random Forest) and **LLMs** (e.g., Llama-3, Gpt-4, XLNet) via **embeddings, prompting, and fine-tuning** to automatically identify solution-related content in developer discussions
 - Designed and implemented **SPRINT** [\[MSR'25\]](#), an LLM-powered GitHub plugin for bug report management; mentored M.Sc. students who studied issue resolution strategies for AI/ML software systems [\[ICSME'26\]](#)
 - Developed and evaluated **uni/multimodal deep learning models** (e.g., SBert, Clip, Blip) for identifying buggy UI screens and components from natural language bug descriptions [\[ISSTA'24\]](#); ran an empirical study on how bug report information affects **LLM agents** in reproducing and fixing UI-centric bugs
 - Designed **dataset construction protocols**, performed large-scale **qualitative analysis** of developer discussions and code changes, and conducted **developer interviews** to characterize real-world issue resolution workflows [\[ICSE'25, ICSE'26\]](#); extended to **multi-document extraction and generation** of code change rationale

Dhaka University of Engineering & Technology, Dhaka, Bangladesh

- **Lecturer** Dec 2018 – Jan 2022
 - Collaborated with the faculty research group to improve the **Kohonen self-organizing map** clustering algorithm for **high-dimensional datasets** [\[IJEECS\]](#)
 - Supervised three undergraduate students on their theses

Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh

- **Undergraduate Student** Mar 2013 – Dec 2017
 - Designed, implemented, and evaluated iris & retina recognition-based multi-modal person authentication system (**undergraduate thesis**) [\[ECCE'19\]](#)

GRANTS

- **NSF Travel Grant** in 2026 to attend ICSE'26.
- **SIGSOFT Travel Support (CAPS)** in 2026 to attend ICSE'26.
- **WM CS Conference Grant** in 2026 to attend ICSE'26.
- **GSA Conference Grant** in 2026 to attend ICSE'26.
- **WM CS Conference Grant** in 2025 to attend ICSE'25.
- **Seed Grant** in 2024 for “Decoding the Issue Resolution Process in Practice via Issue Report Analysis.”
- **SIGSOFT Travel Support (CAPS)** in 2024 to attend ISSTA'24.
- **GSA Conference Grant** in 2024 to attend ISSTA'24.
- **WM CS Conference Grant** in 2024 to attend ISSTA'24.

HONORS & AWARDS

- **International Student Achievement Award** at William & Mary, 2025.
- **Best Poster Award** at W&M CS Symposium for Graduate Studies, 2024.
- **Best Poster Award** at W&M CS Symposium for Graduate Studies, 2023.
- Technical Education Scholarship, Ministry of Education, Bangladesh, 2013–2017.

- Prime Bank Foundation Scholarship, Prime Bank, Bangladesh, 2013–2017.
- Govt. Merit Scholarship, Ministry of Education, Bangladesh, 2010.
- Govt. Merit Scholarship, Ministry of Education, Bangladesh, 2007.
- Govt. Merit Scholarship, Ministry of Education, Bangladesh, 2004.

MENTORING

William & Mary, Virginia, USA

- *Graduate Research Mentor* Jan 2022 – Present
 - Mentored five Ph.D. students from the SEA lab, two M.Sc. students [ICSME'26][MSR'25], and one undergraduate student in various research projects
 - Served as **Graduate International Peer Leader**: supporting new incoming international graduate students at William & Mary

Dhaka University of Engineering & Technology, Dhaka, Bangladesh

- *Undergraduate Thesis Advisor* Dec 2018 – Jan 2022
 - Mentored undergrad students in thesis and competitive programming
 - Advised undergrad students for their mental wellness and academic performance

TEACHING EXPERIENCE

William & Mary, Virginia, USA

- *Graduate Teaching Assistant* Spring 2022–23, Fall 2022
 - Courses: Data Structures, Software Development, Algorithms
 - Conducted office hours, graded assignments and exams, proctored exams

Dhaka University of Engineering & Technology, Dhaka, Bangladesh

- *Lecturer (Instructor of Record)* Dec 2018 – Jan 2022
 - Courses: Structured Programming, OOP, Information Systems Analysis, Human-Computer Interaction, Numerical Methods, Machine Learning
 - Full course ownership: designed syllabi, conducted theory and lab courses, prepared and proctored exams, graded coursework
 - Supervised students in undergraduate research; student advisor for student well-being

Military Institute of Science and Technology, Dhaka, Bangladesh

- *Lecturer (Instructor of Record)* Jan 2018 – Dec 2018
 - Courses: Object-Oriented Programming, Structured Programming, Data Structures, Computer Networks, among others
 - Full course ownership: conducted theory and lab courses, prepared and proctored exams, graded coursework
 - Mentored students in competitive programming

TALKS

Research Paper Presentations

- [ICSME'26] 42nd International Conference on Software Maintenance and Evolution, October 2026, Auckland, New Zealand
- [ICSE'25] 47th IEEE/ACM International Conference on Software Engineering, April 2025, Ottawa, Canada
- [ICPC'25] 33rd IEEE/ACM International Conference on Program Comprehension, April 2025, Ottawa, Canada
- [MSR'25] 22nd IEEE/ACM International Conference on Mining Software Repositories, April 2025, Ottawa, Canada
- [ISSTA'24] 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis, September 2024, Vienna, Austria
- [ECCE'19] International Conference on Electrical, Computer and Communication Engineering, February 2019, Cox's Bazar, Bangladesh

Poster Presentations

- [ICSE'26] 48th IEEE/ACM International Conference on Software Engineering, April 2026, Rio de Janeiro, Brazil
- [MSR'25] 22nd IEEE/ACM International Conference on Mining Software Repositories, April 2025, Ottawa, Canada
- W&M CS Symposium for Graduate Studies, November 2023 & 2024, Virginia, USA

PROFESSIONAL SERVICES	Program Committee Member 23rd IEEE/ACM International Conference on Mining Software Repositories (MSR'26)		
	Conference Session Chair 48th IEEE/ACM International Conference on Software Engineering (ICSE'26)		
PROFESSIONAL AFFILIATIONS	External Reviewer - IEEE/ACM International Conference on Software Engineering (ICSE'26, '24, '23) - IEEE/ACM International Conference on Program Comprehension (ICPC'26) - IEEE International Conference on Software Maintenance and Evolution (ICSME'25, '23, '22) - IEEE International Conference on Technical Debt (TechDebt'26) - ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'23)		
	- Member, IEEE - Member, ACM – SIGSOFT		
ORGANIZING & VOLUNTEERING	- Student volunteer at ICSE'26, ICSE'25, ICSE'24, ESSEC'24, and ICSME'23 - Web chair and student volunteer at W&M CS Symposium for Graduate Studies – 2023 - Student volunteer at W&M CS Symposium for Graduate Studies – 2022 - Student volunteer at Grad Bash – 2023 at William & Mary - Organizer of DUET CSE FEST'19, Dhaka University of Engineering & Technology, Bangladesh - Organizer of CYBERCOMP'18, Military Institute of Science and Technology, Bangladesh		
	Programming: Python, C, C++, Java, C#, Bash, SQL, MATLAB, Android, PHP, Bootstrap, HTML, CSS, $\text{\LaTeX}$ ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, SciPy, Hugging Face, Pandas AI Techniques: LLMs & Transformers (<i>e.g.</i>, Gpt-4, Llama-3, Bert), uni/multimodal models (<i>e.g.</i>, SBert, Clip, Blip), LLM applications (<i>e.g.</i>, embedding, fine-tuning, prompting), multi-modal learning, statistical modeling Software Engineering: Design patterns, architecture and implementation, code review, testing methodologies, refactoring, code quality analysis Tools & IDEs: PyCharm, IntelliJ IDEA, VS Code, Android Studio, Eclipse, CLion, Git		
SKILLS			
REFERENCES	Dr. Oscar Chaparro Assistant Professor Department of Computer Science William & Mary McGlothlin-Street Hall 311, 251 Jamestown Rd. Williamsburg, VA 23185, USA Email: oscarch@wm.edu Phone: (757) 221-2144		
	Dr. Kevin Moran Assistant Professor Department of Computer Science University of Central Florida Room 217A, L3Harris Engineering Center Orlando, FL 32816, USA Email: kpmoran@ucf.edu Phone: (703) 993-6826		
	Dr. Andrian Marcus Professor Department of Computer Science George Mason University Room 4452, Nguyen Engineering Building Fairfax, VA 22030, USA Email: amarcus7@gmu.edu Phone: (703) 993-9237		